ALEXANDER PANCHENKO

C++ Developer

Email: alexander.pancenco@outlook.com
LinkedIn: www.linkedin.com/in/alexander-panchenko-0b4b47414
GitHub repository: https://github.com/Xaliostra
YouTube DEMO: https://youtu.be/cOTKmdtnCmE | https://www.youtube.com/watch?v=k2bw32llpHQ

Professional Summary

C++ / Unreal Engine Game Developer with a Bachelor's degree in Computer Science. Experienced in designing robust game and engine systems from scratch, from custom pathfinding algorithms (JPS) and modular architectures to P2P network prototypes. Passionate about software design, low-level optimization, and solving complex technical challenges with clean, maintainable code.

Technical Skills

- **Programming**: C++, Unreal Engine C++, C# (basic)
- **C++:** STL, smart pointers, RAII, lambdas, object-oriented programming
- **Algorithms:** A*, Jump Point Search (JPS), pathfinding, graph-based algorithms
- **Software Design:** Modular architecture, component-based design, data-driven systems, plugin development
- **Networking:** Peer-to-peer communication (local), multiplayer replication, state synchronization
- **Tools:** Git, Perforce, JetBrains Rider, Visual Studio
- **Debugging & Performance**: Debugging, profiling, performance analysis

Work Experience

Unreal Engine Game Programmer
Delta Design - Unannounced Project
November 2024 - June 2026

• **Contributed** to the development of an unannounced conceptual arcade game on Unreal Engine 5 within a small, agile core team.
• **Integrated** and extended the Gameplay Ability System (GAS) to drive core gameplay mechanics and abilities.
• **Designed** and implemented player movement logic utilizing the Enhanced Input Subsystem.
• **Developed** HUD components and gameplay UI in C++, wiring visual feedback directly to underlying game states and events.

Selected Projects

Tactical Turn-Based Combat System

C++ · Unreal Engine 5 · Algorithms · System Architecture

Independent technical project focused on designing complex gameplay and supporting systems.

- Designed a modular, data-driven C++ architecture using components, gameplay tags, and reusable systems.
- Implemented core systems using Unreal Engine's C++ API and Gameplay Ability System.
- Implemented a standalone replicated inventory plugin using a component-based architecture.
- Implemented **Jump Point Search (JPS)** for grid-based tactical movement.
- Extended the pathfinding system to handle dynamic obstacle updates.
- Developed AI behavior using Behavior Trees and event-driven gameplay logic.
- Designed reusable data-driven abilities and items.
- Developed gameplay UI primarily in C++, with Blueprint integration where appropriate.
- Debugged and profiled systems throughout development to identify correctness and performance issues.

Targeted Limb Combat System

C++ · Unreal Engine 5 · System Design · Gameplay Systems

Independent technical study focused on designing a modular combat system.

- Designed and implemented a C++ combat system using Gameplay Ability System.
- Built a modular melee attack pipeline including combos, animation events, weapon tracing, and dodge mechanics.
- Implemented limb-specific damage calculation using armor, damage multipliers, hit-bone mapping, and gameplay state.
- Developed a tag-driven weapon tracing system that dynamically selects trace sockets based on attack state.
- Implemented screen-space target selection and mapping between selected limbs and gameplay state.
- Developed enemy combat behavior using Behavior Trees and event-driven ability activation.

Blockchain-Based Land Registry

Academic Prototype · C++ · Peer-to-Peer Networking

Bachelor's qualification project developed independently as an academic prototype.

- Designed and implemented a blockchain-based architecture for a land registry application.
- Implemented peer-to-peer communication between nodes in a local network environment.
- Developed transaction validation and block synchronization mechanisms.
- Implemented the application layer for managing cadastral registry data.
- Investigated and implemented the core concepts required for blockchain-based data storage and synchronization.
- Developed the project independently as a bachelor's thesis.

Education

Bachelor of Computer Science

Kyiv National University of Taras Shevchenko - Kyiv, Ukraine

Specialization: Mathematical Informatics

Bachelor's thesis: **Application of Blockchain Technologies in Land Cadastre Registration**